
HaptiConnect: 혼합 현실에서 플레이어 쌍을 위한 공유 햅틱 감지 시스템

신체 착용 햅틱 조끼와 메타 퀘스트 헤드셋을 사용한 게임화된 레이저 미로 데모

HaptiConnect: A Shared Haptic Sensing System for Player Dyads in Mixed Reality

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요약 햅틱 피드백은 몰입형 환경에서 시각만으로는 전달하기 어려운 공간 정보를 전달하는 유망한 채널로 주목받고 있으나, 동일 공간에 위치한 플레이어 간의 대인 공간 관계를 인코딩하는 데 있어서의 가능성은 아직 충분히 연구되지 않았다. 본 논문에서는 두 플레이어가 상체 전방위 360° 진동촉각 피드백을 통해 서로의 상대적 위치를 지속적으로 인식하는 2인 협동 혼합 현실 게임 HaptiConnect를 제안한다. 이 피드백은 '라이프라인'이라는 가상의 공유 연결을 표현하며, 플레이어 간 상호 거리에 따라 동적으로 변화한다. 플레이어는 레이저 미로를 협력하여 탐색하면서 라이프라인을 유지해야 한다. 12명의 참가자(6개 이인조)를 대상으로 시각만, 햅틱만, 시각+햅틱의 3가지 조건에서 피험자 내 파일럿 연구를 진행하였다. NASA-TLX 결과, 시각+햅틱 조건에서 작업 부하가 가장 낮았으며 오류 수도 가장 적었다. 이러한 결과는 지속적인 방향성 진동촉각 피드백을 통한 대인 공간 인코딩이 지각 가능할 뿐만 아니라, 시각 피드백만으로는 달성할 수 없는 방식으로 협동 조율을 능동적으로 지원함을 시사한다.

Abstract Haptic feedback has emerged as a promising channel for conveying spatial information that vision alone cannot reliably communicate in immersive environments, yet its potential to encode interpersonal spatial relationships between co-located players remains largely uninvestigated. We present HaptiConnect, a two-player cooperative MR game in which both users maintain awareness of each other's relative position through continuous 360° vibrotactile feedback around the upper torso, representing a shared virtual connection called the Lifeline. Players must navigate a laser maze while protecting the Lifeline, which changes dynamically based on mutual player distance. We evaluated HaptiConnect using a within-subjects pilot study with 12 participants across 6 dyads and 3 conditions: visual only, haptic only, and visual+haptic. NASA-TLX revealed a clear workload advantage for the combined condition, which also produced the fewest errors. These results suggest that interpersonal spatial encoding through continuous directional vibrotactile feedback is not only perceivable but actively supports cooperative coordination in ways that visual feedback alone cannot.

핵심어: *Mixed reality, haptic feedback, cooperative games, vibrotactile, shared control*

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1. Introduction

Co-located multiplayer games rely on player interdependence and shared control to foster cooperation [1, 3]. While shared control promotes balanced player contributions [2], it also increases coordination demands and can reduce players' perceived competence and autonomy by requiring continuous communication and interpretation of each other's intentions [1].

Haptic feedback is a promising modality for conveying spatial information in immersive environments, yet its use for encoding interpersonal spatial relationships between co-located collaborators remains largely unexplored [5, 7]. Wearable haptic devices such as belts and vests can communicate 360° directional cues and improve spatial awareness, including obstacle avoidance [4, 6]. While prior work has demonstrated haptic communication between remote collaborators through force-based interaction [7], it has not investigated using continuous spatially encoded haptic feedback to support coordination between co-located users.

Therefore, we present HaptiConnect, a two-player cooperative game in which both users maintain awareness of each other's relative position through continuous 360° vibrotactile feedback around the upper torso. Using haptic vests and head-mounted displays (HMDs), we create a virtual connection between players in mixed reality, referred to as a Lifeline. The Lifeline functions as a shared physical constraint, requiring both players to co-manage their mutual spatial relationship, representing a new form of shared control in mixed reality. Players must cooperatively navigate a laser maze. To succeed, players rely on shared spatial understanding and continuous haptic feedback, which dynamically encodes the Lifeline's direction and relative distance based on each player's tracked position.

2. System

HaptiConnect is a co-located two-player mixed reality game in which both participants share the same physical space and wear Meta Quest 3 head-mounted displays to see a digital overlay rendered over the real world. Each player also wears

a bHaptics TactSuit Air haptic vest, enabling full-body vibrotactile feedback distributed across the upper torso. The goal is for both players to cooperatively navigate through a laser maze across three progressively difficult levels, moving from a start zone to an end zone while maintaining a shared virtual connection called the Lifeline. Connectedness is conveyed through two channels: visually, through a line rendered in MR space between the two players that changes color based on their mutual distance; and haptically, through directional vibrations that continuously orient each player toward the other. At close range the Lifeline remains green with a low continuous hum; as distance increases it turns yellow with intermittent vibration; beyond a critical threshold it turns red and emits an intense pulse, indicating it has broken. Contact with laser obstacles also breaks the Lifeline, requiring players to reset to the start of the current level.



Figure 1. Distance of the Lifeline influences color and haptic feedback. In the third image, the Lifeline breaks due to *both* overextension and a collision. The scene depicted is level 1, and the second player (photographer) is not pictured.

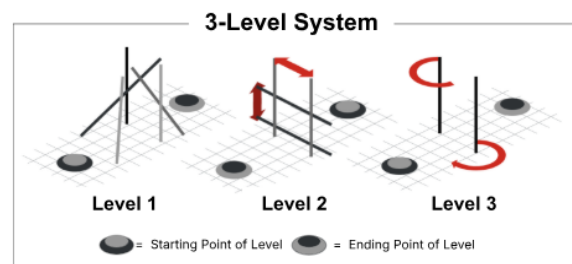


Figure 2. The three-level laser maze configuration. Level 1 presents static lasers, Level 2 introduces two moving sets of parallel lasers, and Level 3 features 2 rotating lasers. Figure made with IcoGrams.

HaptiConnect was implemented in Unity using C# and Unity Netcode for GameObjects for real-time multiplayer state synchronization. Co-location and shared spatial anchors were established via the Meta XR SDK and Meta Building Blocks, which align the coordinate spaces of both headsets to a common real-world origin. Haptic feedback was integrated

using the bHaptics SDK2, which provides directional control over each vest's motor array. Player head positions are synchronized across the network using NetworkVariables, and a LineRenderer component renders the Lifeline between tracked headset transforms. Laser collision is detected via Physics.Raycast against a dedicated Laser physics layer, triggering Lifeline break logic and level reset on contact.

3. Pilot Study

We evaluated HaptiConnect using a within-subjects study with three experimental conditions: visual + haptic (VH), in which both the Lifeline visualization and haptic feedback were enabled; haptic-only (H), in which haptic feedback was preserved but the visual Lifeline was removed; and visual-only (V), in which the Lifeline remained visual but haptic feedback was disabled. The study involved 12 participants aged 18-35, organized into 6 dyads. Each dyad experienced all 3 conditions in fully balanced order. 83.3% of participants had prior experience with HMDs. To control for order effects, condition order was counterbalanced across dyads such that each dyad completed the conditions in a different sequence. Task completion time and error count were recorded per level, and participants completed the NASA-TLX workload questionnaire and stated their condition preference after all conditions were tested.

Completion time was only evaluated for successful attempts, but there was no consistent pattern across conditions (V: M=13.51s, SD=4.15s; H: M=14.38s, SD=9.78s; VH: M=13.56s, SD=5.04s). In most cases, >3 seconds were spent planning the timing aspect of the level, rather than actually moving through it. NASA-TLX revealed a clear workload advantage for the VH condition (V: M=8.61, SD=4.84; H: M=9.71, SD=3.31; VH: M=5.70, SD=1.65), with VH producing the lowest perceived workload in four out of six dyads. The remaining two dyads reported lower workload under the V condition. The V and H conditions produced comparably elevated workload scores overall. 83.3% of participants stated verbally that they favored the VH condition, with the remaining pair (16.7%) preferring V. For least preferred, 50% of pairs rated V as their least favorite, 33.3% rated H,

and 16.7% rated VH.

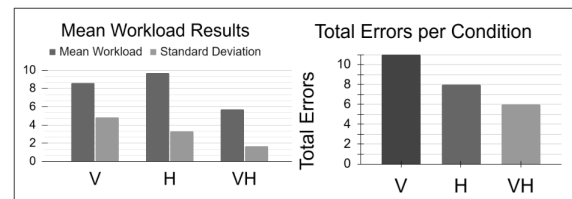


Figure 3. VH caused fewer user errors (Lifeline broken by distance or laser) and a reduced workload.

4. Conclusion

Building on prior work [7], our results suggest that continuous directional vibrotactile feedback supports cooperative coordination beyond visual feedback alone, reducing workload and errors. By continuously encoding interpersonal spatial relationships, the Lifeline fostered shared spatial awareness between co-located collaborators, extending prior haptic interaction research toward cooperative mixed reality games.

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